
Domain Generalization for In-the-Wild Waste Classification: Measuring, Diagnosing, and Mitigating the Gap Between Clean and Real-World Data

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Abstract

Models for waste classification are almost always trained on clean, object-centric photographs, yet they get deployed somewhere much messier: real bins and real landfills, full of clutter, crushed objects, and inconsistent lighting. This project measures that gap directly. Three architectures (ResNet-50, EfficientNet-B3, ViT-Small) were trained on 18,042 images from two clean-domain sources (TrashNet, GD12) and then tested zero-shot on RealWaste, a genuine landfill dataset that never appears in training. The result is a 35–42% drop in balanced accuracy across all three models. Grad-CAM points to why: misclassified images are the ones where the model is attending to background clutter or printed text rather than the material itself. From there we tried three ways of closing the gap. Two of them backfired — heavier synthetic augmentation and a third, more visually diverse training dataset both made out-of-distribution accuracy *worse*. What did work was a smaller, more targeted change: dropping the noisiest class, switching to resize-padding, adding label smoothing, and introducing a border-background augmentation that leaves the object untouched and only varies the surrounding pixels. This pushed zero-shot RealWaste accuracy up to 64.5% for ViT-Small, without the model ever seeing a single target-domain image. We also pushed the approach to its breaking point using TACO, a dataset of litter photographed in natural scenes rather than as isolated objects; accuracy collapsed to 33–39% for every architecture equally, which we read as a task-structure boundary (object-centric classification versus in-context detection) rather than a further generalization failure. A calibration analysis (ECE, AUROC) turned up something we did not expect going in: accuracy and calibration are inversely related across architectures, with ViT-Small being the most accurate model and simultaneously the least trustworthy about its own confidence, while EfficientNet-B3 sits at the opposite end. An independently collected 170-image dataset, photographed by our own team, confirms the mitigation benefit for both CNNs but exposes a limitation specific to ViT-Small. Finally, Wilcoxon and McNemar tests confirm what the large-sample results suggest while appropriately holding back on the smaller-sample comparisons where the evidence is weaker.

1 Introduction

Automated waste sorting is one of the more practical applications of computer vision for recycling infrastructure, at least in principle. In practice, models trained on clean, catalog-style photographs tend to fail once they meet a real waste stream (Babé et al., 2025; Hossen et al., 2024). Part of the reason is well documented in the domain-adaptation literature: there is a gap between the *source domain* used for training (well-lit, single object, uncluttered) and the *target domain* encountered at deployment (crushed, dirty, cluttered, badly lit), and this gap remains a central open problem in applied waste classification (Abdu and Noor, 2022).

Three questions shaped this project. How large is the domain gap, and does it depend on which architecture is used? What visual cues actually cause models to fail out-of-distribution, and can the standard toolbox — more augmentation, more training data — close the gap, or does it need something more targeted? And finally, if the gap can be closed for one kind of real-world data (object-centric photographs), does that success carry over to a structurally different kind of real-world data, such as litter photographed in context rather than isolated?

We address these questions through a systematic experimental pipeline spanning 38 logged experiments (Appendix A), covering baseline training, failure diagnosis via Grad-CAM (Selvaraju et al., 2017), three mitigation strategies, a scope-boundary evaluation, an independently collected validation dataset, calibration analysis, and statistical significance testing.

2 Related Work

Convolutional architectures such as ResNet (He et al., 2016) and EfficientNet (Tan and Le, 2019) remain the dominant backbones for waste classification (Celik, 2025; Sayem et al., 2025), while transformer-based architectures (Dosovitskiy et al., 2021) have seen more limited application in this domain. Babé et al. (2025) specifically studied the generalization ability of foundation models for waste classification and reported substantial accuracy degradation under domain shift, consistent with our findings. Arishi (2025) and Hossen et al. (2024) report strong in-distribution results but do not systematically test OOD generalization or mitigation strategies, which is the primary contribution of this work. We additionally draw on Guo et al. (2017) for calibration methodology (Section 10).

3 Datasets

Training data. We combine two clean-domain sources into a unified 6-class training set (cardboard, glass, metal, paper, plastic, trash), later simplified to 5 classes (Section 7): TrashNet (2,527 images, white-background studio photography) and GD12 (15,515 images, 12 original classes mapped to our 6; battery, biological waste, clothing, and shoes are mapped to *trash* as non-recyclable general waste). This yields 18,042 training images, split 70/15/15 (train/val/test, seed 42).

Out-of-distribution test data. RealWaste (Single et al., 2023) contains 4,752 images of real landfill waste across 9 categories, mapped to our 6 (Miscellaneous Trash, Food Organics, Textile Trash, and Vegetation → trash). RealWaste is held out entirely from training in our primary experiments (Sections 5–6) to serve as a genuine zero-shot benchmark.

Scope-boundary test data. TACO (Proença and Simões, 2020) consists of 1,500 photographs of litter in natural and urban scenes, annotated with 4,784 bounding boxes across 60 fine-grained categories. We map these to our 6 classes and crop each annotation into an individual image to construct a classification-format test set (Section 8).

Own hard dataset. To validate findings on an independently collected source, our team photographed 170 real-world waste items following an object-centric protocol (single item, filling the frame, genuine environmental clutter). The metal class (17 images) was excluded from evaluation after a diagnosed composition failure (small objects photographed against disproportionately large natural backgrounds; Section 9), yielding 142 usable images across 4–5 classes depending on experimental configuration.

We use *balanced accuracy* (mean per-class recall) throughout, as all datasets are class-imbalanced.

Table 1: In-distribution vs. zero-shot out-of-distribution (RealWaste) balanced accuracy, original 6-class setup.

Model	In-Distribution	Zero-Shot OOD	Domain Gap
ResNet-50	95.5%	60.0%	−35.5 pts
EfficientNet-B3	96.0%	53.6%	−42.4 pts
ViT-Small	97.4%	62.2%	−35.2 pts

4 Models and Training

We train three ImageNet-pretrained architectures, chosen to test whether generalization behavior depends on architectural inductive bias: **ResNet-50** (He et al., 2016) (convolutional, local receptive fields, residual connections), **EfficientNet-B3** (Tan and Le, 2019) (compound-scaled convolutional network), and **ViT-Small/16** (Dosovitskiy et al., 2021) (patch-based self-attention, no convolutional locality prior). All models are fine-tuned for 20 epochs using AdamW (initial learning rate 1×10^{-4} , cosine annealing), weighted cross-entropy loss to address class imbalance, and standard ImageNet normalization.

5 Baseline Results: The Domain Gap

All three architectures achieve 95–97% in-distribution balanced accuracy but drop by 35–42 percentage points when evaluated zero-shot on RealWaste (Table 1). ViT-Small generalizes best, EfficientNet-B3 worst; this ranking reproduces consistently across every subsequent zero-shot experiment in this work (Sections 7, 9), suggesting it reflects a genuine architectural property rather than experimental noise.

6 Diagnosis: What Causes the Gap?

We use Grad-CAM (Selvaraju et al., 2017) to visualize which image regions drive each model’s predictions. For ViT-Small, standard Grad-CAM is inapplicable since the architecture lacks convolutional feature maps; we use a patch-resaping transform to recover a spatial attention map from the transformer’s output sequence.

Across all three architectures, correct predictions are associated with activation concentrated on the object’s surface and edges, while misclassifications are frequently associated with activation on background texture or printed text/logos rather than the object itself (Figure 1). This indicates that models trained purely on clean-domain images learn source-domain surface shortcuts (background style, print) rather than generalizable material-level features. We additionally tested whether the heterogeneous *trash* class (aggregating battery, biological waste, clothing, shoes, and other non-recyclables) was a primary driver of the gap by excluding it from scoring; balanced accuracy *decreased*, since trash was consistently the models’ best-performing class on RealWaste. This rules out label heterogeneity as the primary cause and is consistent with the surface-shortcut explanation.

7 Mitigation Strategies

Three mitigation strategies were tried, each one a direct response to what the failure diagnosis above suggested.

Strategy 1: Heavier synthetic augmentation. The first, most obvious idea was to make training harder: standard augmentation was swapped for a heavier pipeline (motion blur, shadows, occlusion, noise) meant to push training images closer to real-world conditions. In-distribution accuracy ticked up slightly (95.5%→97.4% for ResNet-50), but zero-shot RealWaste accuracy went the wrong way (60.0%→57.0%). The likely explanation is that this kind of augmentation just makes the model more confident about the source domain, without teaching it anything that actually transfers.

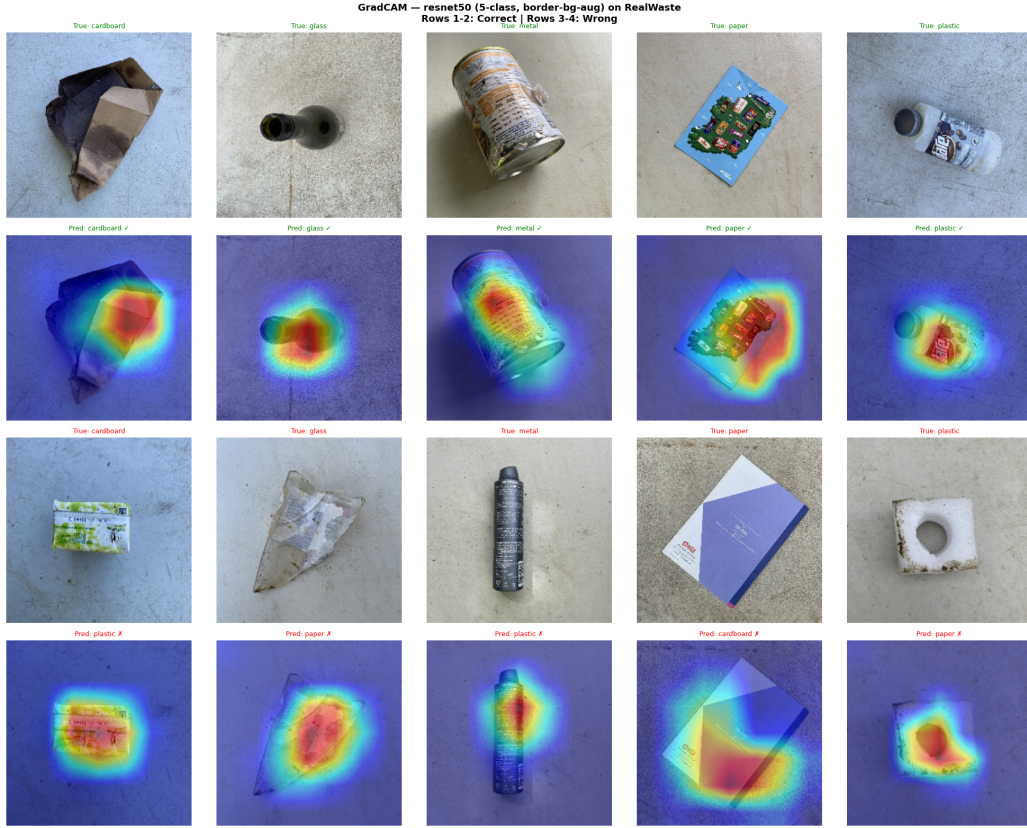


Figure 1: Grad-CAM visualizations for ResNet-50 (5-class, border-background-augmented model) on RealWaste. Top two rows: correct predictions, activation concentrated on the object. Bottom two rows: misclassifications, activation more diffuse or shifted toward surface content.

Strategy 2: Additional same-domain training data. The second idea was more data, specifically a third public dataset (sumn2u, 12,259 images, 10 classes) merged into training. Despite the extra volume and nominally more varied backgrounds, zero-shot accuracy dropped again, to 56.4%. On closer inspection, TrashNet, GD12, and sumn2u all turn out to be visually similar despite coming from different sources — clean, single-object photography, essentially the same domain wearing different clothes. Adding more of that domain just gives the model more source-specific patterns to memorize, not more generalization. That failure is what motivated the redesign below.

Strategy 3 (redesigned pipeline): Border-background augmentation. Following the diagnosis that models over-rely on background cues, we redesign the pipeline with five changes evaluated jointly and then ablated: (i) simplification to 5 classes (cardboard, glass, metal, paper, plastic), removing the heterogeneous trash class entirely; (ii) a resize-and-pad transform replacing center-cropping, to avoid discarding off-center objects; (iii) label smoothing (0.1) and increased weight decay (2×10^{-2}) to reduce overconfidence; (iv) checkpoint selection by validation balanced accuracy rather than validation loss; (v) *border-background augmentation*: rather than segmenting the foreground object (found unreliable; see Appendix B), we replace only the outer border of each training image with a synthetic textured background, leaving the object region untouched. Crucially, RealWaste is not used in training at any point for this strategy—it remains a genuine zero-shot test set throughout.

Table 2 shows a controlled ablation isolating the augmentation’s effect: with every other factor held constant, border-background augmentation improves zero-shot RealWaste accuracy by 1.5 points for ResNet-50—a small but genuine effect, and notably the *opposite* direction from the generic augmentation in Strategy 1, supporting our diagnosis that the type of augmentation matters more than its presence. Rolling this configuration out to all three architectures, ViT-Small achieves 64.5%

Table 2: Ablation of border-background augmentation (ResNet-50, 5-class, all other components held constant), and full rollout to all architectures.

Model	Without Aug. (Control)	With Border-BG Aug.	Δ
ResNet-50	56.5%	58.0%	+1.5 pts
EfficientNet-B3	—	53.8%	—
ViT-Small ($\eta=1e-4$)	—	63.0%	—
ViT-Small ($\eta=2e-5$)	—	64.5%	—

Table 3: Zero-shot balanced accuracy on our independently collected dataset (5-class configuration, metal excluded; 142 images), compared to RealWaste.

Model	RealWaste	Own Dataset
ResNet-50	58.0%	45.2%
EfficientNet-B3	53.8%	41.2%
ViT-Small	63.0%	55.3%

zero-shot accuracy (after a learning-rate sensitivity check reduced η from 1×10^{-4} to 2×10^{-5} ; Appendix D), the best zero-shot result obtained in this work without any target-domain data.

Domain adaptation variant. We separately evaluated a fourth strategy in which 15% of RealWaste (712 images) is added to training, with the remaining 85% (4,040 images) held out for evaluation. This produced substantially larger gains (76.4%, 79.9%, 81.6% for ResNet-50, EfficientNet-B3, and ViT-Small respectively), but we emphasize that this experiment answers a *different research question*: limited-supervision domain adaptation rather than zero-shot generalization. The held-out 85% was never seen in training, but because some RealWaste data informed training, these results are not directly comparable to the zero-shot figures elsewhere in this paper and are reported as a separate, clearly labeled protocol (Appendix C).

8 Scope Boundary: TACO

To test how far the improvements above generalize, we evaluate the border-background-augmented models on TACO, a structurally different type of real-world data. Accuracy collapses to 33–39% across all three architectures, despite the same checkpoints achieving 58–65% on RealWaste. Notably, all three architectures converge to a nearly identical accuracy on TACO (33.3%, 34.1%, 33.8%)—a pattern inconsistent with a generalization failure (which would typically show architecture-dependent variation) and consistent with a task-structure mismatch. We investigated crop dimensions and found 42% of TACO bounding-box crops fall below 100×100 pixels; filtering to larger crops recovers only a small improvement (to 38–39%). We conclude that TACO represents *in-context litter detection* (small, partial objects within complex natural scenes) rather than the *object-centric classification* task our models are trained for, and that closing this gap requires a detection-then-classification architecture (e.g., YOLO) rather than further classifier tuning. We treat this as a documented scope boundary rather than a deficiency of the proposed mitigation.

9 Independent Validation: Own Dataset

To confirm findings are not specific to RealWaste, we evaluate the same border-background-augmented checkpoints on our independently collected dataset (Table 3). The architecture ranking (ViT > ResNet > EfficientNet) reproduces exactly, the fourth consecutive confirmation of this ordering across the project. Absolute accuracy is lower than on RealWaste, consistent with our dataset representing a harder or differently distributed real-world source. The metal class was excluded after diagnosis revealed a photography protocol failure—small objects (e.g., bottle caps, tent pegs) photographed against disproportionately large natural backgrounds, structurally resembling the TACO failure mode—illustrating that even carefully collected validation data is sensitive to composition, and reinforcing the object-centric vs. in-context distinction identified in Section 8.

Table 4: Calibration metrics on RealWaste (OOD). Overconfidence gap = mean confidence – accuracy.

Model	OOD Accuracy	OOD ECE	Overconf. Gap	OOD-Detection AUROC
ResNet-50	60.2%	0.132	+0.131	0.757
EfficientNet-B3	55.4%	0.047	+0.046	0.885
ViT-Small	65.9%	0.153	+0.153	0.666

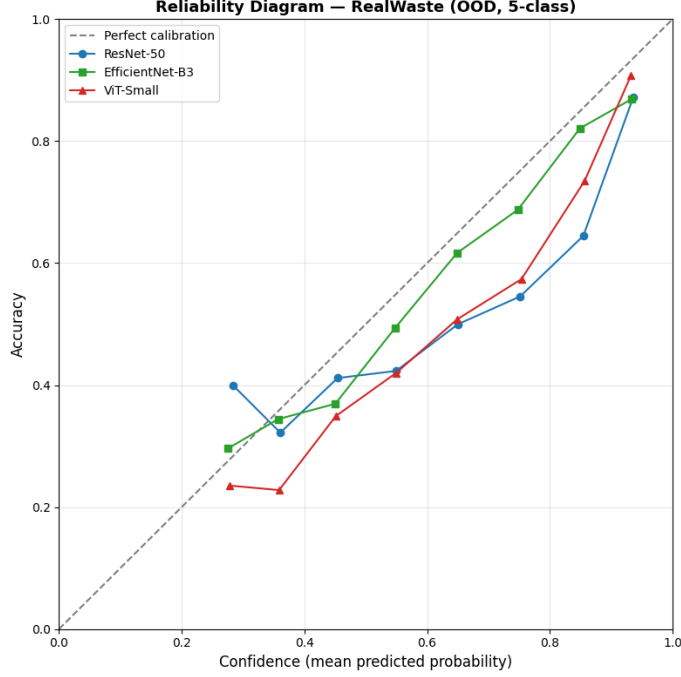


Figure 2: Reliability diagram on RealWaste. All models lie below the diagonal (overconfident) for confidence > 0.4; ViT-Small deviates furthest, EfficientNet-B3 tracks closest to perfect calibration.

10 Calibration Analysis

Following Guo et al. (2017), we compute Expected Calibration Error (ECE) and use max-softmax confidence as an OOD-detection signal (AUROC), evaluating whether models’ confidence estimates are trustworthy under distribution shift.

Table 4 reveals a striking inverse relationship: ViT-Small, the most accurate model, is the *worst* calibrated and least able to signal its own uncertainty (lowest AUROC), while EfficientNet-B3, the least accurate model, is the *best* calibrated. This pattern reproduces on our own dataset (Appendix E), confirming it is a robust architectural property rather than a dataset-specific artifact. The reliability diagram (Figure 2) shows all three models are overconfident in the mid-to-high confidence range (0.3–0.85) but slightly underconfident at low confidence, refining the simple “models are overconfident” narrative. Practically, this suggests EfficientNet-B3 may be preferable in deployment settings where flagging uncertain predictions for human review matters as much as raw accuracy.

11 Statistical Significance

We test pairwise architecture differences using both the Wilcoxon signed-rank test and McNemar’s test (the latter being the statistically appropriate test for paired binary classification outcomes) on per-image correctness. On RealWaste ($n = 3,092$), all pairwise differences are significant ($p < 0.001$) by both tests. On our own dataset ($n = 142$), results are more nuanced: ResNet-50 vs. EfficientNet-B3 is not significant by either test; EfficientNet-B3 vs. ViT-Small is significant by both; ResNet-50 vs.

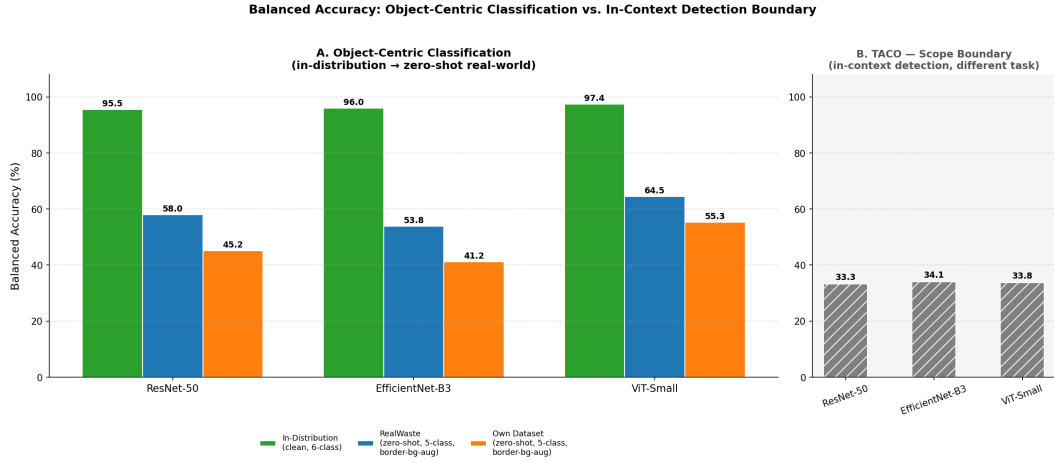


Figure 3: Balanced accuracy across test conditions. Panel A: object-centric classification (in-distribution through zero-shot real-world transfer). Panel B: TACO, shown separately as it represents a structurally different task (in-context detection) rather than a comparable failure.

ViT-Small is significant by Wilcoxon ($p = 0.042$) but *not* by McNemar’s test ($p = 0.057$). We report the more rigorous McNemar result and explicitly note that the own-dataset ViT-vs-ResNet advantage, while directionally consistent with all other results, should be interpreted as suggestive rather than statistically confirmed given the small sample size.

12 Results Summary

Figure 3 summarizes results across all test conditions. We deliberately separate TACO into its own panel to avoid the misleading impression that it represents a fourth point of comparable failure on the same task; as established in Section 8, it measures a different problem entirely.

13 Discussion and Limitations

Our central finding is that *data type matters more than data quantity or synthetic diversity* for closing the domain gap in waste classification: two strategies that added substantial training data (Sections 7, Strategies 1–2) both worsened OOD accuracy, while a targeted, low-cost augmentation change (border-background replacement) produced a small but genuine and statistically-grounded improvement without any target-domain data. The larger gains from limited-supervision domain adaptation (Appendix C) further support this: even a small quantity of genuinely target-domain data outperforms large quantities of same-domain data.

Several limitations warrant statement. First, the domain-adaptation experiment (Appendix C) necessarily uses a portion of RealWaste in training; we have taken care to report this as a separate protocol from our zero-shot results throughout, per methodological feedback received during project supervision. Second, our own dataset (142–170 images) is small, and Section 11 shows that not all architecture comparisons on it reach statistical significance. Third, addressing the TACO scope boundary would require an object-detection architecture, which is outside the scope of this classification-focused project. Fourth, we did not test domain-adversarial training objectives or frozen-backbone fine-tuning, both of which are plausible further mitigation strategies motivated by our findings.

14 Conclusion

We provide a systematic empirical study of domain generalization for waste classification, characterized by five main findings: (1) a large, consistent 35–42% zero-shot domain gap exists across three distinct architectures; (2) common mitigation strategies (heavier augmentation, additional same-domain data) reliably *worsen* this gap; (3) a targeted augmentation addressing the diagnosed

failure mode (background reliance) yields a small, statistically grounded, zero-shot improvement; (4) an accuracy/calibration trade-off exists at the architecture level, reproduced across two independent OOD datasets; and (5) a genuine task-structure scope boundary separates object-centric classification from in-context litter detection. Future work should pursue object-detection-based approaches for in-context litter and explore domain-adversarial training objectives building on the diagnosis established here.

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Declaration of Authorship

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Individual Contributions

Esmail Molapour: data pipeline construction and dataset curation, all model training, baseline and zero-shot evaluation, Grad-CAM diagnosis, design and execution of all mitigation experiments, TACO pipeline and scope-boundary analysis, own-dataset collection protocol and evaluation, calibration analysis, statistical testing, and report writing. **Khaled Ibrahim:** evaluation methodology planning, contribution to calibration metric design. **Khawar Khan:** experiment tracking infrastructure, architecture literature review. **Ashly Varghese:** contribution to early augmentation experiment design.

A Full Experiment Log Summary

This project logged 38 experiments (EXP001–EXP035, with sub-experiments), tracked in `experiment_log.py` in the accompanying code repository. Table 5 summarizes the major experimental phases; full per-experiment notes, hyperparameters, and results are available in the repository.

Table 5: Major experimental phases.

Phase	Experiments
Baseline training & class-weight fix	EXP001–EXP008
Zero-shot OOD baseline (RealWaste)	EXP009–EXP011
Mitigation: augmentation, 3rd dataset	EXP012–EXP015
Trash-class diagnostic	EXP016
Domain adaptation (15% RealWaste)	EXP017–EXP019
TACO scope-boundary evaluation	EXP020–EXP023
Own dataset evaluation & diagnosis	EXP024–EXP027
5-class redesign & border-bg augmentation	EXP028–EXP029
Calibration analysis	EXP030, EXP032
Own dataset re-evaluation (5-class)	EXP031
ViT learning-rate check	EXP033
Statistical significance testing	EXP034–EXP035

B Foreground Segmentation Attempt

An initial attempt at background augmentation used GrabCut-based foreground segmentation to precisely isolate objects before background replacement. Visual inspection revealed this approach frequently misclassified parts of the object itself (particularly light-colored interior regions of open cardboard boxes) as background, corrupting training images. This was identified and discarded prior to any training run, in favor of the safer border-only replacement strategy described in Section 7.

C Domain Adaptation Protocol Detail

We emphasize again that these results answer a different question (limited-supervision domain adaptation) than our zero-shot results and are not directly comparable to them, per feedback received during project supervision regarding evaluation protocol separation.

Table 6: Domain adaptation results: 15% of RealWaste (712 images) added to training, evaluated on the remaining 85% (4,040 images, held out).

Model	Zero-Shot (Table 1)	Domain Adaptation	Δ
ResNet-50	60.0%	76.4%	+16.4 pts
EfficientNet-B3	53.6%	79.9%	+26.3 pts
ViT-Small	62.2%	81.6%	+19.4 pts

D ViT Learning Rate Sensitivity

Motivated by the sensitivity of transformer architectures to learning rate, we re-trained ViT-Small at $\eta = 2 \times 10^{-5}$ (vs. the shared $\eta = 1 \times 10^{-4}$ used for all architectures in Section 7). This improved zero-shot RealWaste accuracy from 63.0% to 64.5%—a modest but genuine gain, indicating the original learning rate was mildly suboptimal for this architecture without overturning the central finding that ViT-Small is the strongest zero-shot generalizer of the three architectures tested.

E Own-Dataset Calibration

Table 7: Calibration metrics on the own dataset (OOD), confirming Table 4.

Model	OOD Accuracy	Overconf. Gap	OOD-Detection AUROC
ResNet-50	43.7%	+0.252	0.873
EfficientNet-B3	36.6%	+0.167	0.923
ViT-Small	54.9%	+0.193	0.790

The AUROC ranking (EfficientNet-B3 > ResNet-50 > ViT-Small) reproduces exactly from RealWaste, confirming the accuracy/calibration trade-off identified in Section 10 is not a RealWaste-specific artifact.

F Confusion Analysis

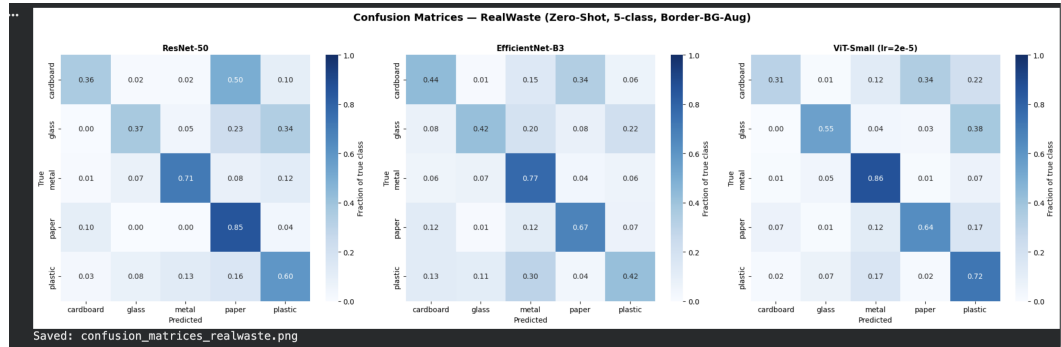


Figure 4: Confusion matrices on RealWaste (5-class, border-background-augmented models). Cardboard→paper and glass→plastic are the dominant, architecture-independent confusions, consistent with genuine material ambiguity in real-world conditions (crushed/wet cardboard resembling paper; broken or clear glass resembling plastic) rather than model-specific weaknesses.